

The Reality of AI Benchmark Measurement: Understanding the Strengths and Limits of AI Evaluation

Series 1, Article 1, April 2026.

by

Ibrahim Sulaiman A.

CEO and Founder, SACS Nig. Limited

As part of the series: “Foundations of AI Evaluation, Research, and Technical Intelligence.”

Abstract

Artificial intelligence (AI) benchmarks have become central to the measurement, comparison, and communication of progress in the field. Benchmark leaderboards have driven scientific innovation, corporate competition, and organizational adoption, and influenced public policy debates. However, despite their pervasiveness and perceived importance, benchmark scores often serve as proxies for unarticulated, aspirational, or contested claims about model capabilities and competencies. This paper investigates the reality of measurement in the age of frontier AI, interrogating the role, relevance, and limitations of benchmarks in evaluating emergent capabilities.

Using a narrative literature review, this study analyzes the scope and power of current benchmarks, the methodological strengths and weaknesses of the measurement paradigm, and the nascent science of benchmark evaluation. The review focuses on the specific competencies being measured by contemporary benchmarks and their relevance to the capabilities of modern foundational models: reasoning, knowledge, software engineering, multimodality, and emerging agentic behaviours. In addition, the review analyzes the theoretical and practical benefits and shortcomings of the benchmarking paradigm, including considerations of standardization, replicability, comparability, and progress in science and technology, as well as ecological validity and generalizability, benchmark contamination, calibration, domain-specificity, and saturation.

Furthermore, this paper investigates trends in benchmark evaluation, identifying the critical role of scientific quality control in fostering confidence in reported scores and supporting evidence-based technological development and societal benefit, and presents the B.E.A.R. (Benchmark Evaluation and Assessment for Responsible interpretations) Framework - a conceptual framework informing the critical assessment of benchmarks. The framework emphasizes five (5) key considerations for those seeking to interpret benchmark evidence: quality, capability, applicability, replication and robustness, and relevance. This study finds that AI benchmarks should be contextualized as scientific measurements, not objective indicators of intelligence or readiness for deployment. As the capabilities of frontier models grow, so should the awareness of the principles informing measurement, interpretation, and application.

Keywords: Artificial Intelligence, AI Benchmarking, AI Evaluation, Foundation Models, Large Language Models, Benchmark Evaluation, Measurement Theory, Responsible AI, B.E.A.R. Framework, Frontier AI.

1. Introduction

Artificial intelligence (AI) is currently undergoing an era of unprecedented capability expansion. This development trajectory has been fueled by foundation models trained at an unprecedented scale and sophistication (Bommasani et al., 2021). Recent advances in large language models (LLMs) and multimodal systems have demonstrated exceptional progress in reasoning, coding, science, and multimodal understandings, with broad implications for both research and practice (Liang et al., 2022; Chang et al., 2024; Stanford HAI, 2025). At the same time, the intensifying competition among AI developers has increased the need for objective assessment measures. In practice, this has meant that benchmark scores have become a de facto symbol of model capability. They dictate technical conversations, inform procurement choices, steer development priorities, and, more generally, act as a focal point of attention for both the academic and popular press (Li et al., 2024; Starace et al., 2025).

At the same time, it is evident that model performance on a given benchmark often has disproportionate impact on the perceived value of that model. If an AI system achieves state-of-the-art results on a particular benchmark, it is often labeled universally superior or more capable, despite the fact that the task captured by the benchmark only represents a small part of the system’s potential applications (Bender & Koller, 2020). This is not necessarily a flaw of the benchmarks. Rather, it is an artifact of how convenient it is to discuss complex, multidimensional, and context-dependent capabilities through a single easily interpretable number. To be clear, such a number is useful, not least because a model’s performance on a given task often does correlate with its overall capability (Bowman & Dahl, 2021). However, it is less obvious how an organization should interpret an AI system’s performance on a benchmark in conjunction with its practical application in the real world. This latter environment is rarely static and well-defined, as opposed to the controlled setting in which a benchmark is typically evaluated (Keller et al., 2026). In practice, this often means that factors such as ambiguity, domain specificity, collaboration, goal alignment, and operationalization all play a role in determining what constitutes “success” for a given AI application but are not captured within a singular benchmark.

Thus, it is not surprising that there has been a recent push to scrutinize the assumptions behind AI benchmarks and, specifically, the implications of their use. A growing body of work highlights the importance of understanding how a given evaluation framework fits within a larger practical application context, as well as the limitations of using particular aggregate scores as a basis for discussion. For instance, BetterBench positions validity, documentation, governance, and maintenance as critical factors in determining the value of a given benchmark (Keller et al., 2026). Similarly, HELM emphasizes the importance of multidimensional evaluation, as opposed to relying on a single metric, as a way to achieve greater practical insight (Liang et al., 2023). More recently, we have seen the appearance of more

challenging evaluations, such as Humanity’s Last Exam, which serve to push the boundaries of frontier models and ensure that their aggregate scores still reflect differential capability (Messick, 1989). Thus, it is becoming evident that the practical assessment of AI systems is a complex and nuanced task that requires significant methodological and conceptual sophistication to truly capture the intricacies of model capabilities. These insights have led to a growing recognition that AI evaluation is not just a matter of choosing the right “scores.” More appositely, it is a scientific and practical pursuit that is concerned with establishing measurement validity and ensuring proper interpretation of evidence.

This article aims to contribute to this discussion by focusing on the phenomenon of AI benchmark measurement from one specific angle. It attempts to highlight the unique position of benchmark scores by emphasizing the ways in which they help assess or misrepresent the state of AI capabilities. Rather than focusing on direct comparison of different models or evaluation frameworks, this article asks what these frameworks say about the systems they are supposed to evaluate. To this end, the paper focuses on questions such as what aspects of AI capabilities are captured by a given score, what aspects are not, and how these factors relate to broader practical considerations. This line of reasoning serves as the basis for a discussion of current trends in AI benchmarking and the associated implications for practitioners and researchers. The present analysis synthesizes the latest insights from the academic literature and the technical reports of relevant benchmarks to identify areas of agreement and potential points of contention.

This article makes two specific contributions to the discussion of AI benchmark measurements. First, it provides a comprehensive, up-to-date overview of the key themes in the field. By doing so, it identifies areas where the research community, practitioners, and policymakers may benefit from a more nuanced understanding of the challenges associated with measurement interpretation. Second, the paper builds upon the insights of recent works (Liang et al., 2023; Keller et al., 2026) to introduce the B.E.A.R. Framework which is a conceptual framework for informed interpretation of AI benchmarks. By focusing on the idea of “responsible interpretation” and emphasizing the need for contextual analysis, this framework aims to support more practical, evidence-based discussions of AI model capabilities. Ultimately, the article is intended to contribute to a deeper understanding of measurement validity within the context of artificial intelligence.

2. Foundations of AI Evaluation

Evaluating artificial intelligence has long been a dynamic pursuit. The methodologies have changed as our capabilities of AI systems grow, and the very definition of evaluation evolves alongside the technology (Russel & Norvig, 2020). Contemporary foundation models represent a qualitatively new challenge compared to their predecessors (Liang et al., 2023). These models operate in multiple domains, interact using natural language, produce images and code, and demonstrate competencies that go beyond the goals

of many of the benchmarks that have been developed for them (Stanford HAI, 2025). Evaluation is now a field of scientific study in its own right, not an ancillary activity to machine learning (Bommasani et al., 2021; Keller et al., 2026).

At a general level, evaluation is concerned with assessing an AI system’s degree of success in accomplishing particular objectives or performing particular tasks or activities. Beyond merely identifying the best model, evaluation provides critical insights into reliability, generalizability, safety, efficiency, fairness, and applicability in a given context (Sculley et al., 2015; Keller et al., 2026). These insights have downstream implications for science, engineering, regulation, and commerce (Yeung, 2020). The very importance of evaluation has risen due to the enhanced role that AI plays in societies. An erroneous statement of evaluation can have wide-ranging effects, from misleading the scientific community to damaging business, policymaking, healthcare, national security, and judicial systems (Raji et al., 2020). Given this context, there is a need to understand the role of benchmarks in evaluation, the principles that should guide them, and their importance to understanding AI progress.

A benchmark is a standardized evaluation tool, typically comprising a set of instructions, datasets, tasks, performance metrics, and scoring mechanisms that allow for consistent measurement of a model’s proficiency in performing well-defined activities or achieving particular objectives (Reuel et al., 2024). At a fundamental level, benchmarks are critical to ensuring the validity and reliability of evaluation results, as well as enabling comparisons between different approaches at a given point in time or across different moments (Liang et al., 2022; Wang et al., 2019). Like any standardized test in psychology, education, or physics, a benchmark is necessary to provide meaningful information about the performance of a model (Hendrycks et al., 2021). By design, benchmarks define the boundaries of AI performance measurement, and so they shape the direction of scientific inquiry and innovation (Glaser, 1963; Keller et al., 2026). The very existence of the benchmarks influences what is possible in AI.

The rise of the concept of the benchmarks is closely linked to the evolution of the field of AI itself. The early days of symbolic AI were dominated by attempts to encode logical constraints, and evaluation was focused on the system’s ability to follow these constraints correctly (Russell & Norvig 2020). The machine learning paradigm shifted the emphasis on the model’s ability to generalize from training sets to test sets. The rise of deep learning saw the development of increasingly sophisticated benchmarks that fuelled rapid progress in computer vision, audio analysis, and language understanding (LeCun et al., 2015). The advent of foundation models and frontier AI has added new dimensions to the capabilities that researchers seek to measure through benchmarks (Junior et al., 2024). These models can perform competently in a variety of domains, and so evaluating their progress requires consideration of an extended set of tasks that go beyond mere language modeling (Wang et al., 2019; Bommasani et al., 2021). As a result, today, there is a proliferation of benchmark categories that attempt to measure diverse competencies of large language

models and evaluate their progress towards broader capabilities, including coding, reasoning, scientific knowledge, multimodality and, more recently, agentic behaviour (Li et al., 2024; Wang et al., 2019; Mirzadeh et al., 2025; Yu et al., 2024; Li et al., 2024b).

However, these competing benchmark families represent diverse approaches to the problem of progress measurement, and none of them can capture the entire distribution of abilities that modern foundation models possess (Li et al., 2024; Keller et al., 2026). A number of benchmarks are devoted to measuring knowledge retention and verbal reasoning (Wang et al., 2024), whereas others focus on mathematical reasoning (Mirzadeh et al., 2025), or software engineering (Yu et al., 2024). A subset of benchmarks tackle science-specific competencies, and a few address the challenge of multimodality (Li et al., 2024b). More recently, several benchmarks have emerged that attempt to measure the extent to which AI systems can act as agents in complex environments (Shen et al., 2024; Ling et al., 2025; Harkar, 2026). Overall, these diverse benchmarks provide an opportunity to obtain a more comprehensive understanding of the capabilities of modern AI systems, but they highlight the complexity of attempting to characterize such systems using any one particular set of evaluations.

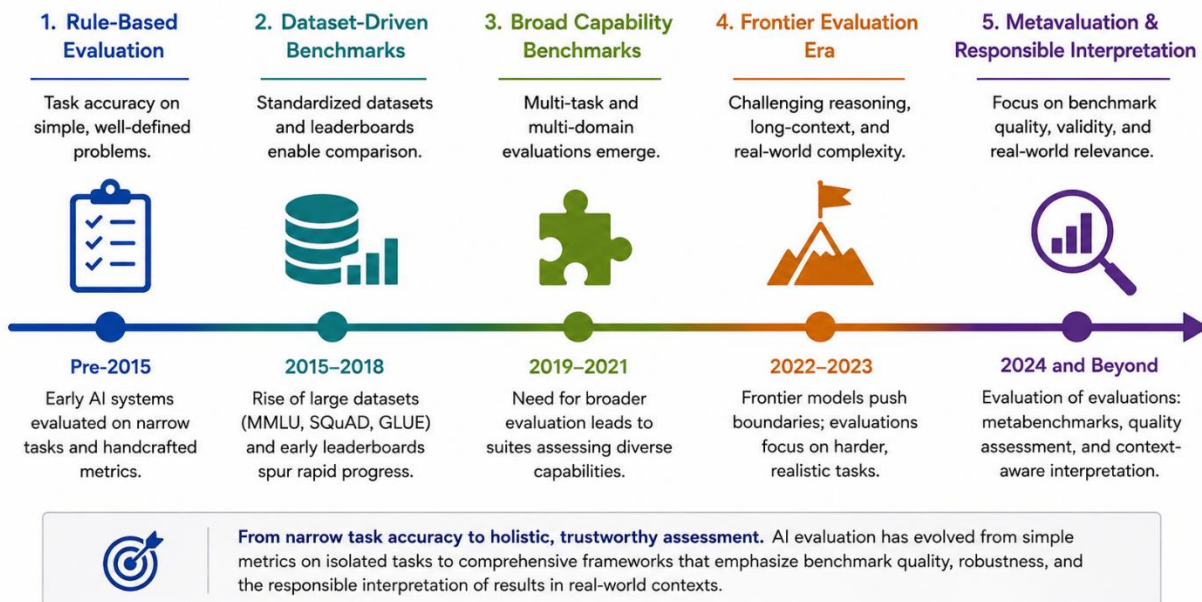


Figure 1. Evolution of AI evaluation.

The variety of benchmarks supports an important idea that is often ignored when comparing the power of different artificial intelligences. The ability of AI is multi-faceted (Liang et al., 2022). Reasoning capacity, creativity, factual knowledge, adaptability, safety, collaboration, and real-world applications are just several areas in which one model will outperform another (Shukla, 2025). Each benchmark highlights unique

aspects of performance that were identified, tested, scored, and defined differently (Reuel et al, 2024). As such, understanding what a given score means and how these scores relate to one another requires nuance and context (McGregor et al., 2026). For instance, proficiency in the form of a mathematical reasoning benchmark only confirms a model’s capability in this particular area. It does not imply that the same model possesses equal levels of expertise in fields such as cybersecurity, science, and policy analysis (Kiela et al., 2021).

With this idea in mind, it becomes evident why benchmarks are needed to evaluate and compare the performance of modern AI models. On the one hand, they help transform vague statements about capabilities into concrete metrics. In another sense, benchmarks bring order to the chaos of model development and enable standardized performance assessment. In short, models are compared to one another based on a person’s ability to complete specific tasks. Therefore, it is always necessary to distinguish between these tasks and the real world. This is the key point that differentiates the concept of AI performance from conventional practice. The following section will discuss the specific performance indicators that are used in contemporary benchmarks to identify and analyze the strengths and weaknesses of modern artificial intelligence. These ideas will become the basis for the next few assignments, which will focus on a more detailed discussion of the limitations of the existing measurements (Keller et al., 2026).

3. What AI Benchmarks actually measure

The ubiquity of benchmark leaderboards creates an illusion that AI systems are ranked according to a single integrated metric. They are not. Modern benchmarks encompass a sprawling and growing ecosystem of tasks, and each targets a particular set of skills (Srivastava et al., 2023). More importantly, the very concept of benchmarks is limited in what it can represent: a score on a particular task only reflects proficiency in the skills that were operationalized when designing the task. A math reasoning benchmark will rarely provide reliable insights about scientific creativity, and a software engineering benchmark offers little information about multilingual dialogue, policy analysis, and so on (Bommasani et al., 2021; Chang et al., 2024).

Most benchmarks are based on the operationalization paradigm: high-level concepts are narrowed down to particular sets of measurable skills. Researchers identify a certain ability they wish to investigate, design representative tasks with objective scoring, collect training and test data, and define how the model answers will be compared to the reference solution or behavior. The process always involves an element of reduction since, by definition, operationalization replaces abstract qualities with concrete measurements (Bommasani et al., 2021; Keller et al., 2026).

Knowledge-based benchmarks represent one of the largest and most established categories of benchmarks. They reflect the model’s aptitude to perform tasks that require factual knowledge and general cultural

competence. In particular, models are tested on their ability to answer questions from various domains, spanning history, law, medicine, economics, and the sciences (Center for AI Safety, 2026). For example, the MMLU benchmark comprises dozens of domains and utilizes multiple-choice questions to assess the model's level of knowledge (Hendrycks et al., 2021; Reuel et al., 2024). In turn, this characteristic is frequently associated with the breadth of training data and the presence of domain-specific knowledge (Hendrycks et al., 2021). However, performance on this type of benchmarks rarely speaks about the presence or the depth of understanding.

Reasoning-based benchmarks are closely related to the knowledge-based ones but focus more on the model's ability to perform logical deductions, mathematical computations, and commonsense reasoning. The presence of these skills is often inferred from the quality of factual knowledge (Davis, 2023; Suzgun et al., 2023). Unlike knowledge-based assessments, reasoning-based ones attempt to capture the reasoning process itself, making them more informative about the internal workings of the model. For instance, GSM8K and MATH benchmarks are centered around mathematical problem solving and require the models to demonstrate step-by-step solutions (Cobbe et al., 2021; Hendrycks et al., 2021). The rising interest in such tasks reflects a paradigm shift in AI research: the field is moving from accumulating factual knowledge to utilizing it in complex manipulations with symbols and information.

Another important category of benchmarks is related to programming skills. Modern language models are often expected to generate working software, reason about its design and implementation, locate and resolve defects, and describe the behavior of existing programs. Different tasks are captured by HumanEval, MBPP, and SWE-Bench, among other assessments (Chen et al., 2021; Jimenez et al., 2024). Each of these benchmarks operationalizes a specific component of programming ability, from implementing programs from natural language specifications to analyzing and repairing real software repositories. Software-related assessments have been at the forefront of ML research for years, and one of the reasons for this is the ability to quantitatively evaluate correctness through automated testing.

Modern systems need to be able to reason about a mixture of modalities in a single interaction with the user (Masry et al., 2022). As a result, the corresponding benchmarks focus on multimodality, capturing the ability to process and generate text, images, graphics, sounds, and video. For example, the MMMU benchmark evaluates language models' aptitude to answer questions that involve reasoning over text and images (Lu et al., 2024). Similar tasks are used to probe the model's understanding of figures, graphics, illustrations, and diagrams (Lu et al., 2022). The ability to perform multimodal reasoning is becoming increasingly important as the majority of human communication uses multiple modes simultaneously (Chang et al., 2024). The value of such assessments is further enhanced by the fact that the use of multiple modes usually implies richer interaction and greater cognitive engagement (Lu et al., 2022).

A wide variety of emerging benchmarks targets increasingly specific abilities. Scientific reasoning benchmarks operationalize the model’s aptitude to perform tasks that require advanced scientific knowledge (Rein et al., 2023; Zhao et al., 2026). Long-context benchmarks reflect the model’s ability to effectively process and reason about very long documents (Kuratov et al., 2024; Yu et al., 2025). Agentic benchmarks highlight the model’s ability to operate various tools and perform extended reasoning with limited supervision (Jimenez et al., 2024; Wei et al., 2026). Finally, PaperBench and Humanity’s Last Exam are extremely challenging benchmarks designed to probe the most advanced models (Starace et al., 2025; Keller et al., 2026; Phan et al., 2025).

Table 1. Major AI Benchmarks and Their Primary Evaluation Domains

GLUE	NLP understanding	Language
SuperGLUE	Advanced language understanding	Language
MMLU	General knowledge	Knowledge
ARC	Scientific reasoning	Knowledge
GSM8K	Arithmetic reasoning	Mathematics
MATH	Competition mathematics	Mathematics
GPQA	Graduate scientific reasoning	Science
HumanEval	Code generation	Programming
MBPP	Python programming	Programming
SWE-Bench	Repository engineering	Software engineering
MMMU	Multimodal reasoning	Vision-language
MathVista	Visual mathematical reasoning	Multimodal
ChartQA	Chart understanding	Multimodal
LongBench	Long-context reasoning	Memory
AgentBench	Autonomous agents	Agentic AI
WebArena	Web navigation	Agentic AI
PaperBench	Scientific research	Research
Humanity's Last Exam	Frontier evaluation	General capability

Although these benchmark families seem different, they have some similarities. These families include standardized data sets, standardized evaluation protocols, quantitative performance metrics, and standardized testing environments (Winfield & Studley, 2024). Therefore, it becomes possible to compare different models developed by different researchers using similar conditions. In addition, it becomes possible to analyze trends by working with different generations of models (Keller et al., 2026).

However, the variety of categories under which AI models are tested is also notable. In fact, it is this variety that makes one realize that the capabilities of AI cannot be summarized by a single metric (Solanke, 2026).

Each category reflects a particular perspective on the capabilities of AI models. The entire set of these categories is much more informative than the sum of its parts, although it still remains a kind of abstraction of real-world performance (Wang et al., 2019). Thus, understanding what these benchmarks represent is only part of the challenge. No less important is the need to understand why these specific metrics are used to evaluate AI models and what advantages they provide despite their limitations (Bommasani et al., 2021; Keller et al., 2026).

Table 2. Common Evaluation Metrics

Metric	Typical Benchmarks	Purpose
Accuracy	MMLU, GSM8K	Percentage of correct answers
Exact Match (EM)	QA benchmarks	Exact string agreement
F1 Score	SQuAD, QA tasks	Balance of precision and recall
pass@k	HumanEval	Probability that at least one of k generated programs passes all tests
Success Rate	SWE-Bench, AgentBench	Completion of end-to-end tasks
Win Rate	Arena-style evaluations	Preference-based comparison
Calibration/Error	Safety and reliability benchmarks	Confidence vs. correctness

4. The Strengths of AI benchmarking

Though frequently criticized, AI benchmarking has proved itself to be among the most important innovations in the field of artificial intelligence in recent years (Junior, 2024; Curcio, 2025). Almost every recent innovation has been associated with benchmark comparisons proving the superiority of the new models over the existing ones (Li et al., 2024b). It was no coincidence. Progress in science is always based on measurements, and measurements need standards to compare various results in a fair and transparent way (Liu et al., 2026). Such standards are provided by AI benchmarks. Without them, claims about better models would be mostly based on subjective demonstrations or hard-to-check evaluations (Li et al., 2024a). The primary strength of benchmarking comes from its ability to standardization. Academics from various institutions, countries, and firms will be able to analyze the models with the help of identical datasets, score functions, and performance measures (Liang et al., 2022). This standard methodology will facilitate the comparison of results and make interpretations of the reported advancements less ambiguous (Winfield & Studley, 2024). Benchmarking such as MMLU, for example, provides an equivalent of reference point that can be used to compare performances within different areas of science. Even though there is no ideal

benchmark, benchmarking makes possible to set a scientific basis on which it is much easier to identify advancements (Reuel et al., 2024).

Standardization also facilitates reproducibility, which is considered an essential part of scientific research (Sierra-Herrera et al., 2026). It is necessary to compare competing theories using the same methodologies and obtain roughly similar results. Reproducibility is important in terms of building confidence in the findings and spotting problems in the methodology or its application. In the field of artificial intelligence where training processes and computational capabilities differ significantly from one organization to another, benchmarking becomes a means of achieving some methodological consistency. This way, benchmarking does not only serve as a tool for testing but as a way to enhance scientific credibility (Morishige & Koshihara, 2025).

Benchmarking has further sped up technological advancements through the definition of measurable research goals (Rein et al., 2023). Advancements that were once only described in qualitative terms can now be quantified, thus allowing scientists to observe progress and note where models fail to perform well (Reuel et al., 2024). This has fostered a kind of innovation loop. Weak performance on one particular benchmark frequently stimulates the development of improved architectures, training methods, or inference algorithms, which then get tested on the same or similar benchmarks.

Comparability is yet another significant contribution. There are great differences between artificial intelligence models in terms of architecture, number of parameters, training data, optimization approaches, and purposes. It is impossible to make any comparisons without common evaluation tasks (McIntosh et al., 2025). Benchmarks provide a controlled setting that allows studying the differences (Solanke, 2026). This is especially useful for those researchers who cannot conduct their own tests in the same conditions as the developers of the models. Benchmarking provides some level of transparency, which allows for the evaluation of the performance of the models.

The utility of benchmarking goes far beyond academic studies. The importance of benchmarks is growing in organizations when they choose AI systems to deploy, distribute research budgets, or assess market products (Reuel et al., 2024). Benchmark performance becomes a useful measure for developers to pinpoint flaws in the models they develop. For investors, benchmark outcomes become an indicator of technological progress (Lu et al., 2026). Increasingly often, policymakers and regulators cite benchmarking results when estimating possible social effects of innovative AI technologies (Li et al., 2024b). In all these cases, the results of benchmarking are used as a kind of supporting evidence rather than proof of anything definite (NIST, 2023). This can be explained by the lack of alternatives that could provide similar standardized assessment of evolving models.

Also, recent advancements in AI benchmarking demonstrate how benchmarking technology keeps evolving (Stanford HAI, 2026). The introduction of the BetterBench framework puts the focus on evaluating the

benchmarks rather than evaluating the model performance per se. This framework includes parameters such as the level of documentation, transparency, maintainability, and methodological soundness (Reuel et al., 2024). This is definitely another step forward for the field of evaluation science. It emphasizes the idea that any benchmark score will be meaningful only if the benchmark itself is scientifically reliable. In addition, the case of Humanity's Last Exam is another example of the evolution of benchmark design in response to advancements in AI capabilities (Phan et al., 2025). The new benchmark does not include well-known datasets but instead features advanced questions created specifically to ensure discriminative ability for the next-generation models (Center for AI Safety, 2026). All the examples prove the following idea: AI benchmarking is not a static collection of datasets. This is a dynamic process that always evolves.

Perhaps, one of the best aspects about AI benchmarking is the possibility to convert abstract notions into something tangible and measurable (Yeung, 2020). Words like "reasoning", "knowledge", "coding skills", "scientific aptitude" or other similar words cannot be measured directly through observations alone, and benchmarks translate those into a measurable task with an observable result. The result may never be perfect, but at least gives us the means to discuss our progress more effectively and consistently.

None of the aforementioned strengths mean that the benchmark score is something that needs to be taken on face value without questioning (Jimenez et al., 2024). Instruments of measurement are useful because they measure certain phenomena in certain conditions, but do not describe everything that exists in reality (Stanford HAI, 2026). This needs to be kept in mind always. Benchmarks will continue to remain indispensable to AI research because of their ability to give standardized, repeatable and scalable measures of model performance. However, their importance hinges on knowing what exactly does that measurement represent. This becomes even more apparent once the performance on the benchmarks is compared to the practical application of AI technology.

5. The Reality: What benchmarks do not measure

The value of benchmark tests for AI is hard to deny. The same goes for their drawbacks. Benchmark scores becoming the lingua franca of comparisons between state-of-the-art models have resulted in their being open to misinterpretations (Liang et al., 2022). A model exceeding all other models in performance in one or more benchmarking environments is claimed to be overall "better," while such a claim has very little to do with the available evidence (Chang et al., 2024). Benchmarks are testing tools, not a representation of intelligence (Keller et al., 2026). The results of benchmarking are valid within the scope of the test design. But the problems arise once these limits are forgotten (Bommasani et al., 2021).

The most important limitation is ecological validity. In general, the benchmarks used to assess models operate within well-controlled conditions where the nature of the task is known, the data sets are curated in advance, and the success criteria are also determined (Kane, 2013). The real-world environment is much

more forgiving for people. Tasks that come to scientists, security professionals, legal experts, or policy makers do not usually come in neat packages that offer easy answers. These tasks include gaps in information, inconsistent goals, uncertain assumptions, and shifting constraints. While the benchmark might be able to test if a particular model is capable of solving the task or answering the question in a mathematics or science context, it says little about how it would cope in a long-term investigation (Reuel et al., 2024).

This difference becomes even more critical in light of professional workflows. Take for example a scenario where an AI helps a cybersecurity analyst while investigating an incident response case. It is not just about data retrieval or code production. It encompasses hypothesis creation, correlating data, thinking from an adversary's perspective, documenting, communicating with colleagues, and reassessment based on new data (Liu et al., 2025). Existing benchmarks test bits and pieces of this process (Kehkashan et al., 2026). They rarely test the entire process as a coherent cognitive activity. High benchmarks' scores hence demonstrate that an AI has acquired some valuable skills but not that it knows how to function.

Another limitation relates to domain expertise. A lot of benchmarks use questions that are based on specific fields of science or standard exams, giving an illusion that a good model demonstrates expertise at the same level (Khalafyan, 2025). While this statement is partly true, there are some qualifications to make. Domain expertise goes beyond knowing correct answers to questions. Professional expertise involves professional decision-making, contextual thinking, tacit knowledge accumulated in practice, moral reasoning, as well as understanding when the provided information is not enough to draw conclusions (Phan et al., 2025). All these traits are hard to measure in a benchmark environment. Thus, benchmark success cannot be considered professional expertise in medicine, engineering, cybersecurity, and science (Bommasani et al., 2021).

Reasoning poses a similar problem. Modern frontier systems have shown impressive success in benchmarks related to mathematics and logic especially with the use of highly advanced inference methods (Rueda et al., 2025). However, there is still a controversy around what such benchmarks actually tell us about reasoning. The correct answer can be achieved either via real multi-step inference, statistical pattern recognition skills, memorizing similar examples or their combination (Wei et al., 2022; Zhang et al., 2022). This is important from the point of view of benchmarking because such benchmarks usually test only observable output and not the process that produced this output. Thus, a good result shows that a system was able to complete a task successfully (Kim et al., 2023). It does not mean anything about how it did it and if it will perform well under the different circumstances (Glazer et al., 2024).

Calibration constitutes yet another significant but frequently neglected limitation (Guo et al., 2017). Leaderboards of benchmark tests usually stress accuracy without considering much confidence estimation (Minderer et al., 2021). A situation in which the model makes right decisions in 90 percent of cases does

not mean that it recognizes when it is going to make mistakes. In decision-making contexts, overconfidence might do more harm than mistakes themselves (Aliferis & Simon, 2024). Diagnostics in medicine, financial calculations, intelligence work, and computer forensics imply that the uncertainty should be evaluated together with correctness. The latest research related to "Humanity's Last Exam" demonstrates the need to take confidence estimation into account since even high-performing models can make mistakes being very confident about it (Phan et al., 2025).

The connection between benchmarks and generalization requires thought, too (Ito et al., 2025). Benchmarks always work with a finite data sample. But models have been used in open-ended contexts where new problems arise, evolving language and new combinations of information occur (Koh et al., 2021). It's good when a model does well on a benchmark because then you know that it can perform well on the distribution of evaluation. Whether or not it will be able to apply its skills elsewhere is another empirical matter (Miyai et al., 2024). This question has gained particular relevance as researchers study distribution shifts, domain adaptation, and model robustness. The more alike the environment and benchmark are, the better.

Also relevant is the matter of benchmark contamination (Balloccu et al., 2024). Contemporary language models are trained on enormous amounts of data gathered from a variety of public and private sources (Mundlamuri et al., 2025). The identification of whether benchmark questions, answers, or any information related to them can be found in the training dataset can be quite hard. In case of an overlap between an evaluation dataset and the training dataset, it is possible that the reported performance will be inflated and the real generalization ability overestimated. This problem has inspired the growing trend toward the development of benchmarks resistant to contamination, hidden evaluation datasets, and continuous test sets (Golchin & Surdeanu, 2025).

A whole other challenge arises due to the incredible speed at which AI is progressing (Mohajan, 2025). Eventually, successful benchmarks themselves will become obsolete. Because as frontier models get better, tasks which used to be challenging become unable to differentiate between them (Phan et al., 2025). Scores on the test start to converge toward the maximum score achievable by that particular benchmark, making comparison harder and harder (Pandit, 2026). This effect is known as the ceiling effect by measurement scientists (Menzies et al., 2008). The benchmark itself is not scientifically obsolete. However, its measurement capability decreased due to the fact that the target population is simply too far ahead compared to when the benchmark was created. The creation of "Humanity's Last Exam" shows how it happens. Instead of being obsolete due to poor design, the benchmark was created to solve a different challenge: to differentiate between frontier models (Phan et al., 2025).

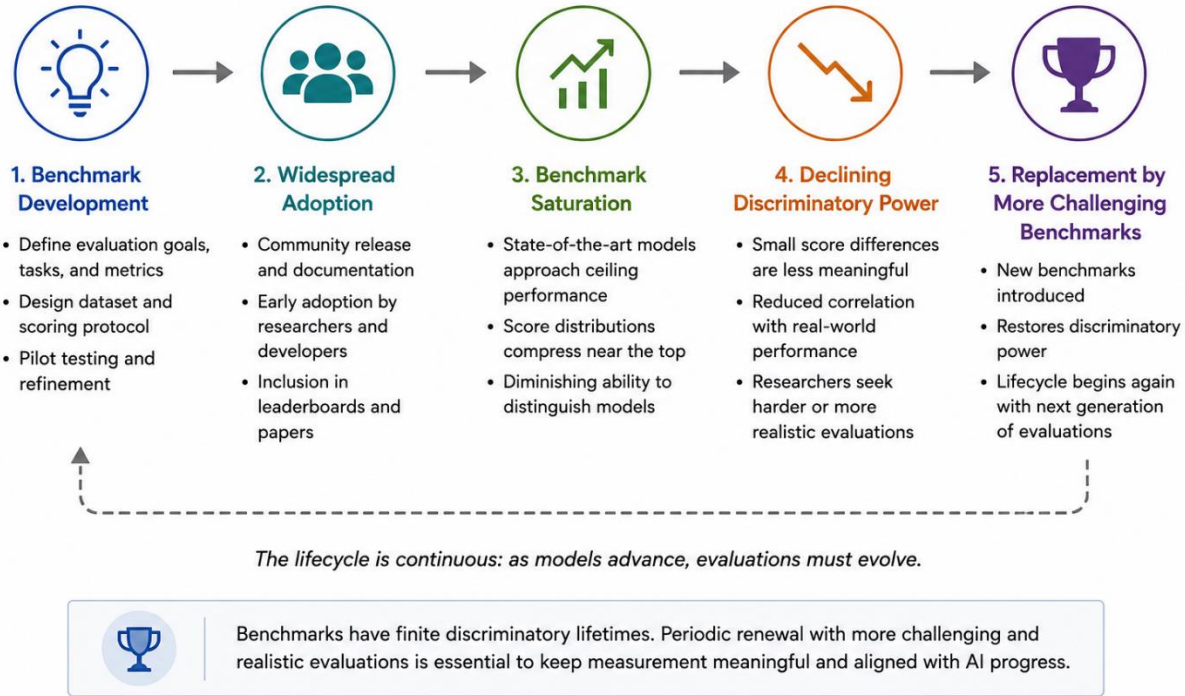


Figure 2. Benchmark Lifecycle in Frontier AI Evaluation

There are other factors that are not captured in benchmark leaderboards, such as economic and operational factors. Different models can obtain similar benchmark results, but have vastly different computational costs, latencies, energy usage, deployment complexity, or access, among other factors (Cheng et al., 2025). These factors directly impact adoption in many organisations. The improvement in benchmark accuracy at a marginal cost could be of limited practical value if the increase is matched by significantly greater inference costs or infrastructure requirements. In a similar way, security, privacy, explainability, regulatory compliance, and integration to other workflows are still key issues for enterprise deployment, but are comparatively less covered in most of the benchmark suites (Reuel et al., 2024).

The most basic constraint, perhaps, is conceptual. Intelligence isn't one thing that can be measured. It includes components of reasoning, memory, learning, creativity, adaptation, communication, judgment, collaboration, and many more aspects, all of which are still objects of ongoing research (Glazer et al., 2024). Specific benchmarks are used to isolate specific constructs for objective measurement. That is a scientific necessity of simplification. It is also necessarily incomplete: A demand is put on the instrument of measurement that it was never designed to meet when a single test or even an entire benchmark suite is expected to measure the entire complexity of intelligent behaviour (McIntosh et al., 2025).

An awareness of these cautions doesn't lessen the significance of benchmarking. Rather, it outlines its proper place in the evaluation of AI. Benchmark scores are important pieces of evidence. They're not

definite assessments of intelligence or ability or readiness for deployment. These can be understood responsibly only if attention is paid to the quality of the benchmarks as well as to model performance. As scientific instruments, the instruments should be the subject of the same critical scrutiny as is applied to the systems being evaluated. This change is more and more reflected in recent research on AI evaluation, and is the core of the next section.

6. Evaluating the benchmarks

So far, most of the discussion has been about the benchmarks that shed light on the capabilities of AI systems (Bommasani et al., 2021). The almost inevitable question is then raised. How can we evaluate if the benchmark is trustworthy? Compared to model evaluation, this question has been much under-researched, partly because benchmarks were regarded as being neutral measures of student performance and were therefore not considered as a research artefact in their own right that ought to be subjected to scrutiny (Liang et al., 2022). Such an assumption is rapidly proving to be hard to justify. A benchmark is not just a bunch of questions. It represents methodological decisions regarding intelligence, measurement of performance, tasks representative of intelligence and forms of evidence sufficient for it (Reuel et al., 2024). Those decisions affect all the consequences that ensue.

This different perspective has engendered a new direction for AI research, which can be summarized as benchmark evaluation, the study of evaluating the quality of evaluation instruments, rather than the evaluation of models. (Reuel et al., 2024). It's not a big difference, but it's a significant difference. It is possible for a poorly designed benchmark to give highly reproducible results but measure the wrong construct (Thomas & Uminsky, 2020). Similarly, a benchmark can become obsolete due to changes in the capabilities of AI, becoming less effective over time in the differentiation of more capable systems (Keller et al., 2026). In both instances, confidence in benchmark scores is based on confidence in the methodological integrity of the benchmark.

This can be explored using measurement theory (Messick, 1989; Prokopchina, 2024). The essential properties of an evaluation instrument that can be found in various fields of science including psychology, education and engineering are that the instrument has to possess (Cronbach & Meehl, 1955; Salaudeen et al. 2025). The most popular are validity, reliability and reproducibility (Messick, 1989). The principles outlined above are equally applicable to AI benchmarking, but it's only recently that these concepts have come to the forefront in the field of machine learning research.

Validity is related to the question of whether a benchmark measures what it purports to measure. (Kane, 2013) This may be the most basic set of criteria for any assessment tool (Glaser. 1963). For mathematical reasoning to be measured, for instance in a benchmark, high scores ought to be based on mathematical thinking, not on superficial pattern matching or memorization (Ling et al., 2025). Validating may not always

be simple, as complex cognitive functions are rarely directly observable. Instead, researchers make inferences about them based on performance on carefully designed tasks (Messick, 1989). It is crucial for benchmark design and construction, scoring approach, and empirical validation to ensure the quality of such inferences (Liu et al., 2026).

Reliability is about a different type of question. Is the benchmark consistent, when other similar models are tested under similar circumstances? (Nunnally & Bernstein, 1994). When the benchmark is not reliable, it adds to the variability in evaluation which makes meaningful comparison hard (Curcio, 2025). With the difference in reported performance between frontier models decreasing, consistency is all the more important. If the measurement process is not sufficiently stable, small improvement is scientifically meaningful only if the process is stable (Pandit, 2026).

Relatedly, there is the issue of reproducibility, which involves the ability to obtain similar results if other researchers use the same benchmark and evaluation methods (Pineau et al., 2021). Reproducibility has become a major issue in computational research across various fields and AI is no different (Morishige & Koshihara, 2025). Reproducible science is facilitated by the documentation of standards or benchmarks, publicly available datasets (where it makes sense to do so), clearly defined evaluation procedures, and accessible implementation details (Gundersen & Kjensmo, 2018). The lack of these elements makes it harder to substantiate the results of the benchmark scores and diminishes their usefulness as evidence (Sculley et al., 2015).

This discussion has been greatly extended by recent research. The BetterBench framework is one of the most systematic attempts to assess AI benchmarks systematically instead of viewing them as gospel, as is commonly done (Reuel et al., 2024). Rather than evaluating the performance of models, BetterBench evaluates benchmarks throughout their entire lifecycle, including documentation, transparency, governance, statistical reporting, maintenance, dataset construction, implementation consistency, and sustainability. The framework states that "benchmark quality must be measured by clear methodological criteria, not by its popularity or widespread use. That's a significant conceptual shift. Familiarity should not be taken as evidence to build scientific confidence (Cheng et al., 2025).

A key finding that has come from this work is that benchmark quality is not just about how the benchmark is implemented. Governance also matters. Benchmarks need to be continually maintained, periodically revised, documented, monitored for contamination, and have processes for fixing weaknesses. (NIST, 2023) Frontier AI benchmarks build on the assumption that there are rapidly-changing research environments instead of static datasets. Questions that are effectively different from state-of-the-art models a few years ago could become trivial in the wake of subsequent model architecture and training improvements (Mundlamuri et al., 2025). Evaluation thus is not a one-time publication, but an ongoing process. This observation naturally leads to the notion of the benchmark lifecycle.

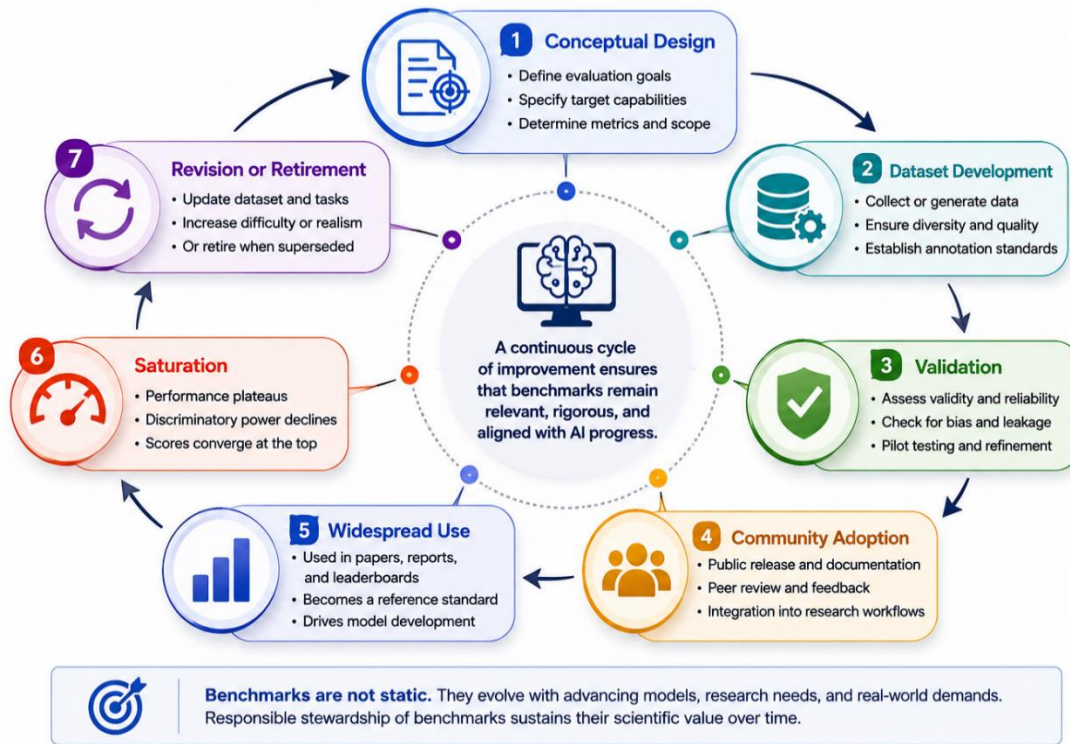


Figure 3. Lifecycle of an AI Benchmark

The lifecycle highlights the limited time when benchmarks have maximum utility. In their early stages, they offer good discriminatory power as few systems perform exceptionally well. Models slowly improve as research takes place until scores on the benchmark are around the top of the assessment range (Stolovitzky et al., 2023). After that, the benchmark's discrimination of meaningful capability differences starts to decrease. The benchmark does not automatically mean that the school has failed. Instead, the population that it was meant to assess has shifted. New benchmarks are then created and the cycle repeats. Humanity's Last Exam is a modern example of this process of evolution and a direct response to the saturation that was observed in several widely-used benchmark suites (Zhai et al., 2026).

Another aspect that is gaining growing importance is the quality of the data sets. Clearly, any benchmark datasets will make decisions on sampling strategy, domain coverage, linguistic diversity, cultural representation, annotation quality, and difficulty calibration (Gebru et al., 2021). These decisions can affect fairness and generalizability. Skewed data sets might systematically support a specific way of reasoning, educational experience, or linguistic structure. Narrow datasets might overestimate performance because they don't capture every type of complexity that occurs in real-time application scenarios (Bender & Friedman, 2018). Therefore, qualities of benchmark should be evaluated in two ways: statistical properties and representativeness of data on which it is based (Stoyanovich & Howe, 2019).

It's also crucial to maintain transparency. The researchers need to know: how were the benchmark questions created, who created them, what assumptions were used in developing the datasets, how were the annotations validated, and what are the remaining caveats in the creation of the datasets (Jiang et al., 2026). Comprehensive documentation is needed to ensure the proper interpretation of benchmark scores by future research as well as to signal when contexts warrant cautious interpretation of the results of evaluation (Reuel et al., 2024b). Transparency also promotes scientific accountability by making methodological choices explicit and not implicit (Chmielinski et al., 2024).

These factors remind us of a major change in the way that benchmark evidence should be understood. It is important to keep in mind that scores obtained on a benchmark should not be judged separately from the benchmarks that generate them. Rather, model performance confidence should be expected to be related to the confidence in the evaluation instrument (Reuel et al., 2024). Benchmarks with low quality documentation and/or methodology warrant careful interpretation. On the other hand, significant increases on less reliable and soundly designed evaluation frameworks (LEFs) might not yield as strong scientific evidence as smaller improvements on more demanding and clearly designed LEFs (Liang et al., 2022).

It's an important difference, but not a significant one. The evaluation process of AI does not start with the scores of the models. It starts with the quality of the measurement system itself (Messick, 1989). Understanding this relationship offers the underpinning for a more systematic way of interpreting evidence from the benchmarks, one that takes into account both the nature of the benchmark itself and the situation in which the evidence from the benchmark will be used in the real world ultimately for decision making purposes. The succeeding section further enriches this view with a structured approach to the responsible interpretation of AI benchmark results: B.E.A.R. Framework.

Table 3. Forms of Validity Relevant to AI Benchmarks

Type of validity	Meaning	AI benchmark example
Construct validity	Does the benchmark measure the intended capability?	Does MMLU truly measure knowledge rather than memorization?
Content validity	Are the benchmark tasks representative of the domain?	Does a cybersecurity benchmark reflect real investigative work?
Criterion validity	Does benchmark performance predict real-world success?	Do SWE-Bench scores correlate with software engineering productivity?
Ecological validity	Do benchmark conditions resemble operational environments?	Does an agent benchmark approximate real workplace tasks?

7. The B.E.A.R. Framework: A Framework for Responsible Interpretation of AI Benchmarks

The above-mentioned discussion has made two inter-related points evident. The first one being that benchmark scores can be useful for making claims regarding certain AI capabilities in controlled circumstances. The second point that follows from this is that the scores will have meaning to the extent that the benchmark on which they are based is scientifically credible. The above-mentioned points make clear that there is more to AI evaluation than numerical comparisons.

While many works have proposed techniques for creating or verifying benchmarks, very few works have looked into the issue of benchmark result interpretation after publishing. This problem is getting more important by the day. Nowadays, companies base their decisions regarding procurement on benchmark leaderboards, scientists justify their claims based on benchmarks, investors consider benchmarks when making assessments of AI companies, and decision-makers turn to benchmarks in conversations about AI regulation (Li et al., 2024b; McGregor et al., 2026). The same benchmark performance may then affect technical, business, and regulatory decisions at once, although the contexts for each decision are different. On the basis of recent developments in the holistic evaluation of artificial intelligence (Liang et al., 2023), benchmark evaluation (Keller et al., 2026), and measurement validity (Messick, 1989), this paper presents a “Benchmark Evaluation and Assessment for Responsible Interpretation (B.E.A.R.) Framework.” The framework does not seek to displace current methodologies but to enhance their use through a series of questions that need to be considered when interpreting benchmark evidence. It is not the aim of the framework to decide which model is better than another. Instead, it seeks to help evaluators think about what benchmark evidence really shows, what is still uncertain, and if the evidence is suitable for the situation at hand.

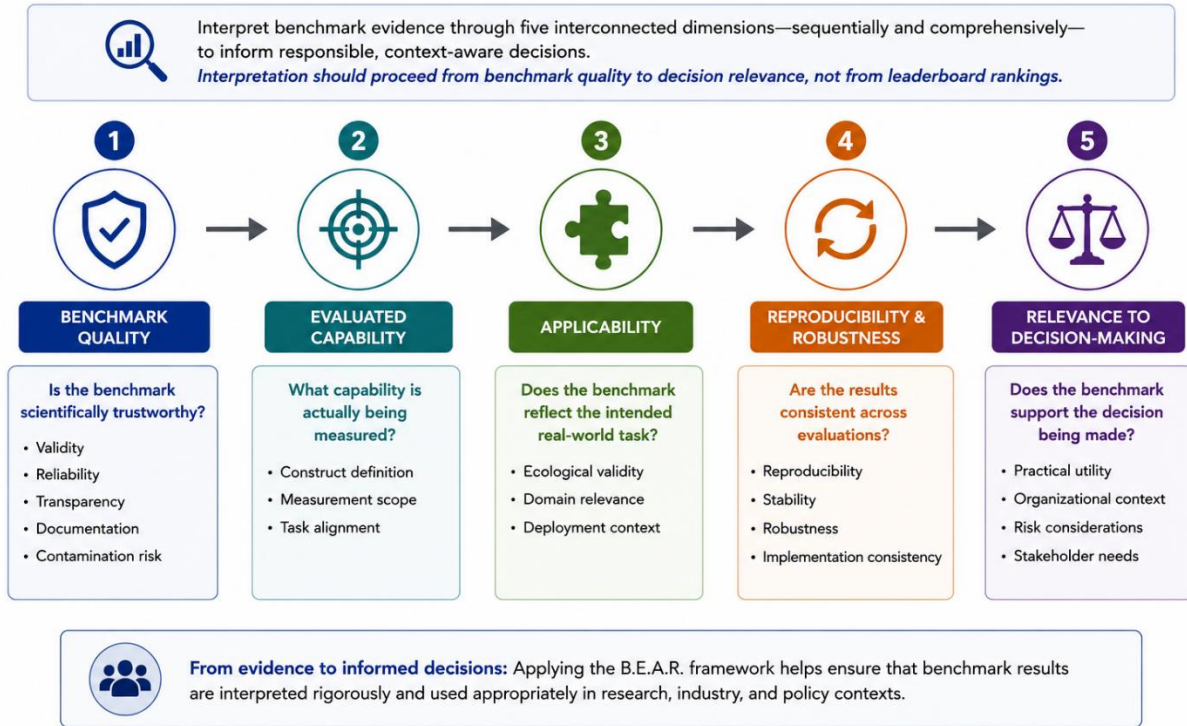


Figure 4. The B.E.A.R. Framework for Responsible Interpretation of AI Benchmarks

The framework consists of five complementary dimensions.

- **Benchmark Quality**

Interpretation starts with the benchmark itself. It is essential to review whether the benchmark itself has adequate methodological rigor before assessing the models' performances. Issues related to validity, reliability, reproducibility, documentation, dataset quality, transparency, contamination potential, and maintainability must first be considered (Messick, 1989; Salaudeen et al., 2025). A benchmark with poor methodological rigor does not constitute strong evidence despite its good performance scores. It is simply an application of an elementary concept taken from measurement theory. The credibility of findings cannot surpass the credibility of the tool used for measurement.

- **Evaluated Capability**

Secondly, there is an issue of determining which ability is being assessed. This seems like a straightforward issue but tends to be forgotten about. The benchmarks apply to specific constructs and not general intelligence. For example, the benchmark for coding assesses the ability to code under certain circumstances. The benchmark for mathematical reasoning assesses structured quantitative reasoning. The multimodal benchmark assesses the integration of information in different modalities. There is some overlap among these abilities, but they should not be equated.

- **Applicability**

Just because an AI benchmark performs well, it does not guarantee good performance in all deployment scenarios. The issue of applicability is more about the degree of similitude of the benchmark to the AI system’s practical use context. An AI benchmark that performs very well at graduate level math may not display the same abilities when the AI is tasked with legal drafting, intelligence analysis, or cybersecurity investigations. Hence, applicability therefore bridges the gap between laboratory evaluation and practical deployment.

- Reproducibility and Robustness

Benchmark results should be consistent for repeated runs within the same evaluation environment, and even when implemented using different versions of models that are not significantly different in design or behaviour (Sierra-Herrera et al., 2026). Beyond just being reproducible (or test-retest reliability), this concept addresses the resilience of results to different conditions. Models that perform consistently across diverse benchmark families, different evaluation environments and independently different implementations are more credible than those that appear strong on the basis of a narrow test condition. Similarly, test-based findings supported by replication studies have more weight than single scores from a company’s private leaderboard.

- Relevance to Decision-Making

This last dimension asks arguably the most crucial question. Is the benchmark providing evidence that is relevant and appropriate to the decision at hand? Each decision is made within a context. Choosing an AI assistant for reviewing scientific literature will call for different evidence than assessing a model for financial forecasting or digital forensic analysis. Decision makers should be wary of conflating good benchmark scores with general applicability. The benchmark evidence must be viewed in light of operational requirements, legal or regulatory restrictions, security considerations, the computational expense of deploying the product, the requirement for human oversight and explanations. Benchmark performance would thus serve as one part of a more comprehensive evidence set.

Table 4. The B.E.A.R. Framework for Responsible Benchmark Interpretation

Framework Dimension	Guiding Question	Primary
Benchmark Quality	Is the benchmark scientifically trustworthy?	Validity, reliability, transparency, documentation
Evaluated Capability	What capability is actually being measured?	Construct definition and measurement scope
Applicability	Does the benchmark reflect the intended real-world task?	Ecological validity and deployment context

Reproducibility	&	Are the results consistent across	Replication, stability, and robustness
Robustness		evaluations?	
Relevance to Decision-Making		Does the benchmark support the decision being made?	Practical utility, organizational context, and risk

While the power of the B.E.A.R. Framework comes from the combination of its parts, the consideration of each one individually can be misleading. A benchmark, though scientifically rigorous, might not measure the capabilities an organization requires. And a scientifically relevant benchmark that appears perfectly suited to an organization’s needs might be poorly documented or not easily reproducible. To avoid misinterpretation, the components of an effective benchmark must be considered together, rather than choosing the one that appeals most to a favoured conclusion.

Another aspect that makes B.E.A.R. model a powerful tool is that it promotes a different way of looking at things. Leaderboards are inherently competitive as they convert the evaluation of a problem into a ranking of numerical scores. Yet, any scientific evaluation needs an interpretation prior to any comparative analysis. Meaningful benchmark results come from an assessment of benchmark quality, evaluated ability, purpose of evaluation, reliability of findings, and context of decision-making all together. The B.E.A.R. Framework is not presented as a definitive solution to AI evaluation. The field continues to evolve alongside the systems it seeks to measure. Nevertheless, the framework offers a structured approach for navigating an increasingly complex evaluation landscape, one in which understanding what benchmark results mean has become almost as important as obtaining the results themselves.

8. Discussion

Based on the foregoing analysis, it appears that AI benchmarking is now at the threshold of a paradigm shift. Over the years, the primary concern in evaluating AI systems was whether they were getting better. Benchmark leaderboards have been answering this question through the provision of consistent evidence of improvement in each generation of these systems (Stanford HAI, 2025). However, in recent times, things are not quite the same anymore. Advanced AI systems are evolving quickly, benchmark ecosystems are growing, and evaluations themselves have become scientific subjects of study. There is a gradual shift away from evaluating AI systems toward assessing the quality of the evaluations themselves (Liang et al., 2022). The shift is consistent with a general rule in the field of measurement science. There should never be any separation between an assessment tool and the underlying assumptions inherent in the design (Kane, 2013). Academic tests assess certain types of knowledge and not the whole of intelligence. Clinical assessments give an idea of the likelihood of a disease and do not necessarily determine the presence of the disease.

Economic indicators tell what is good about the national performance but at the same time, they fail to show other things. The same should be said about AI benchmark tests.

This means that benchmark literacy has to become an indispensable research skill. Having benchmark scores is not enough anymore; researchers have to understand benchmark construction, underlying assumptions, data representativeness, and applicability of results (Reuel et al., 2024; 2024b). There is a subtle, yet fundamental change. Scientific competence now requires understanding not only the model outputs but also the measurement processes behind them.

For the AI developers, this viewpoint can motivate a wider understanding of model optimization. Optimization based on benchmark scores alone can result in some improvements but, at the same time, overlook other aspects of performance that cannot be easily measured with standard benchmarks (Li et al., 2024a). Reliability, uncertainty quantification, cooperation with humans, adaptability to novel situations, and safe operation are under-represented in benchmarking ecosystem despite being critical in actual operational context. Any improvement on a leaderboard, therefore, should not be an end goal itself. Rather, benchmark performance needs to be considered alongside other measures of model quality (McIntosh et al., 2025; Lambert, 2025).

This debate applies just as much to companies considering AI solutions as well. Benchmark results have become increasingly influential in procurement decisions as they offer an easy-to-use comparison tool. But convenience must never be mistaken for comprehensiveness. A firm deploying AI for cybersecurity investigations, clinical decision support, financial analyses, or even scientific research must consider a variety of factors that go way beyond benchmarking accuracy. Often enough, integration capabilities, computational efficiency, privacy considerations, compliance issues, explainability, maintenance needs, and human supervision will prove to be much more significant in determining success than any small variations in benchmark performance.

Implications for public policy and AI governance can be drawn from the findings. There is an increasing reliance on benchmark evidence for AI capability, risk, and national competitiveness by government agencies and regulators (NIST, 2023). This trend will undoubtedly continue as more advanced AI systems are incorporated into critical infrastructures and public services (Elliot, 2019). However, benchmark results are not enough on which regulatory decisions are based. Evaluation approaches are needed that address broader issues of robustness, transparency, accountability, safety, and societal impacts (Harkar, 2026). While benchmarking can support AI governance, it cannot replace governance altogether. It is important to make this distinction since there is a tendency to see performance and responsible use of AI systems as synonymous in public debates.

Additional findings can be made regarding recent innovations like BetterBench and Humanity's Last Exam. The use of such initiatives demonstrates that AI evaluation has become increasingly reflexive (Reuel et al.,

2024; Phan et al., 2025). Apart from testing how effective benchmarks are in differentiating models, researchers started questioning whether benchmarks adhere to basic scientific principles of measurement. The described trend represents another step toward the maturity of AI as a research area. Mature sciences often assess the adequacy of their measuring tools in addition to the phenomena measured. Therefore, it seems that AI evaluation is also going in that direction.

The B.E.A.R. Framework that has been developed in this paper should be seen against the background of this trend. The proposed framework does not aim at replacing the existing benchmark evaluation techniques or setting up a new general standard for evaluations. Instead, it provides an interpretive perspective to be used in analyzing the benchmark evidence prior to its usage in making scientific, business, and policy decisions. The importance of this approach is that it encourages evaluators to go beyond the leaderboard positions and pay attention to the qualities of benchmarks, evaluated capabilities, applicability, reproducibility, and decision relevancy.

Looking back now, one realization that stands out is that the assessment of AI has become increasingly challenging because of the increased capability of the technology. The simplicity of the benchmarks was justified by the fact that the models demonstrated clear capabilities and differences in their performance. This is no longer the case. Modern AI systems operate in many different fields, communicate via different interfaces, and are increasingly becoming involved in processes which include humans, other software and even data from the outside world. In order to assess the systems, it is required to use approaches which take into account their complexity, instead of simplifying it to a number. The benchmarks will still have their place in such processes. Their most valuable contribution, however, may turn out not to be the ranking itself, but the discussion which is based on the evidence they provide.

9. Limitations and future Research

There are some limitations in this study that need to be taken into consideration when interpreting the findings of this study. Firstly, it is a narrative review and analytical synthesis and not an empirical evaluation. The arguments in this paper are therefore based on literature and documentation of benchmarks, not on actual benchmark experiments. This method allows for a wide-ranging focus on the evaluation of AI in the present, but it does not bring empirical evidence to support the proposed B.E.A.R. Framework or make causal analyzes between the interpretation of the benchmark and the quality of the decision (Snyder, 2019; Webster & Watson, 2002).

Second, AI benchmarking is a very dynamic field. There is a remarkable rate of change in frontier models, benchmark suites and evaluation methodologies. New benchmarks are added on a regular basis, old ones updated and hard assessments can get saturated in no time at all, as models get better and better. It is

important, therefore, to consider the benchmark examples used in this article as illustrations of the methodological trends rather than as permanent benchmarks.

The B.E.A.R. Framework is also a theoretical element that has yet to be put into practice with empirical applications. While combining elements of measurement science, AI evaluation, and responsible AI governance, its applicability has not been proven in various domains and organizational contexts. Future studies should test the framework in, for example, cyber security, scientific research, software engineering, healthcare, and public administration to establish its effectiveness in improving the interpretation and use of benchmark evidence.

In addition to the validation of the framework, there are several research opportunities that could be pursued. The research areas of benchmark governance, contamination detection, uncertainty calibration and evaluation of agentic AI systems, long-horizon reasoning, and collaborative human-AI workflows warrant further research. Significantly more focus should be placed on incorporating benchmark evidence alongside practical factors like security, explainability, regulatory compliance and operational risk. Enhancing benchmark quality, interpretation, and governance will be as significant as enhancing benchmark performance as AI systems continue to play a larger role in critical societal and economic services.

10. Conclusion

AI benchmarks are essential for today's AI research. They offer a consistent and repeatable way to assess more and more competent models and have allowed researchers to quantify the shifts in skills in areas like reasoning, language understanding, software engineering, and multimodal learning. In addition to research, benchmark results now shape the competition in the commercial marketplace, organizational AI adoption, public policy and regulatory debate. These are now no longer just technical evaluations but are becoming an increasing factor in decisions with important scientific, economic and societal implications.

This review will propose to see benchmark scores as scientific evidence rather than in terms of intelligence or real-world performance. Each benchmark tests a particular set of skills in a specific set of circumstances and each benchmark has methodological limitations which restrict what can be reasonably concluded from its outcomes. Achievement at a high level gives clear evidence of achievement of the assessed construct, but is not necessarily taken as evidence of general capability, operational readiness or as a representation of domain knowledge. Frontier AI systems are getting better and more impactful as they advance, and knowing these differences is becoming ever more critical.

The review also showcases the methodological shift on which AI evaluation is currently in transition. New projects like BetterBench as well as more stringent benchmark suites like Humanity's Last Exam are an indication of the justifiable evolution of the importance of assessing the quality of the benchmark, alongside the system being evaluated. AI benchmarking is thus becoming a field beyond the scope of model

evaluation, to also include the validity, transparency, reproducibility, and responsible interpretation of measurement.

In this context, this article presents the Benchmark Evaluation and Assessment for Responsible Interpretation (B.E.A.R.) Framework. The framework does not call for another benchmark, but offers a framework for interpreting the evidence of a benchmark by using five complementary dimensions: benchmark quality, evaluated capability, applicability, reproducibility and robustness, and relevance to decision making. The framework invites evaluators to think about these aspects as a whole, rather than focusing on leaderboard scores, helping to emphasize more the interpretation of evidence in a realistic way of the using of AI in real life situations.

The future of AI assessment, however, will be more than just increasingly advanced benchmarks. It will be determined by the quality of measurement tools, the fidelity with which the benchmark evidence is applied, and the discernment of those who apply the benchmark evidence to make decisions for research, organization, and public policy. Benchmark scores should, therefore, not be considered statements of scientific fact, but rather as information to be used as part of the scientific judgment. To ensure that the next generation of AI development and governance continues to have a credible and trustworthy benchmark, strengthening benchmark literacy together with the design of the benchmark will be key.

References

1. Aliferis, C., & Simon, G. (2024). Overfitting, underfitting and general model overconfidence and under-performance pitfalls and best practices in machine learning and AI. *Artificial intelligence and machine learning in health care and medical sciences: Best practices and pitfalls*, 477-524.
2. Balloccu, S., Schmidtová, P., Lango, M., & Dušek, O. (2024). Leak, cheat, repeat: Data contamination and evaluation malpractices in closed-source LLMs. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 67-93).
3. Bender, E. M., & Friedman, B. (2018). Data statements for natural language processing: Toward mitigating system bias and enabling better science. *Transactions of the Association for Computational Linguistics*, 6, 587-604.
4. Bender, E. M., & Koller, A. (2020). Climbing towards NLU: On meaning, form, and understanding in the age of data. In *Proceedings of the 58th annual meeting of the association for computational linguistics* (pp. 5185-5198).
5. Bommasani, R., Hudson, D. A., Adeli, E., Altman, R., Arora, S., von Arx, S., Bernstein, M. S., Bohg, J., Bosselut, A., Brunskill, E., Brynjolfsson, E., et al. (2021). On the opportunities and risks of foundation models. Stanford Center for Research on Foundation Models. <https://arxiv.org/abs/2108.07258>
6. Bowman, S., & Dahl, G. (2021). What will it take to fix benchmarking in natural language understanding?. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies* (pp. 4843-4855).
7. Center for AI Safety. (2026). A benchmark of expert-level academic questions to assess AI capabilities. *Nature*, 649(8099), 1139-1146.
8. Chang, Y., Wang, X., Wang, J., Wu, Y., Zhu, K., Chen, H., Yi, X., Wang, C., Wang, Y., Ye, W., Zhang, Y., et al. (2024). A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology*. <https://arxiv.org/abs/2307.03109>
9. Chen, M., Tworek, J., Jun, H., Yuan, Q., Pinto, H. P., Kaplan, J., Edwards, H., Burda, Y., Joseph, N., Brockman, G., Ray, A., et al. (2021). Evaluating large language models trained on code. <https://arxiv.org/abs/2107.03374>
10. Cheng, Z., Wahnig, S., Gupta, R., Alam, S., Abdullahi, T., Ribeiro, J. A., ... & Wolff, R. (2025). Benchmarking is Broken--Don't Let AI be its Own Judge. *arXiv preprint arXiv:2510.07575*.
11. Chmielinski, K., Newman, S., Kranzinger, C. N., Hind, M., Vaughan, J. W., Mitchell, M., ... & Chang, A. (2024). The CLeAR documentation framework for AI transparency. Harvard Kennedy School Shorenstein Center Discussion Paper.

12. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., & Schulman, J. (2021). Training verifiers to solve math word problems. <https://arxiv.org/abs/2110.14168>
13. Curcio, E. (2025). Benchmarking Reasoning Reliability in Artificial Intelligence Models for Energy-System Analysis. arXiv preprint arXiv:2510.19836.
14. Elliott, A. (2019). The culture of AI: Everyday life and the digital revolution. Routledge.
15. Gebru, T., Morgenstern, J., Vecchione, B., Vaughan, J. W., Wallach, H., Iii, H. D., & Crawford, K. (2021). Datasheets for datasets. *Communications of the ACM*, 64(12), 86-92.
16. Glaser, R. (1963). Instructional technology and the measurement of learning outcomes: Some questions. *American Psychologist*, 18(8), 519–521. <https://doi.org/10.1037/h0049294>
17. Glazer, E., Erdil, E., Besiroglu, T., Chicharro, D., Chen, E., Gunning, A., ... & Wildon, M. (2024). Frontiermath: A benchmark for evaluating advanced mathematical reasoning in ai. arXiv preprint arXiv:2411.04872.
18. Golchin, S., & Surdeanu, M. (2025). Data contamination quiz: A tool to detect and estimate contamination in large language models. *Transactions of the Association for Computational Linguistics*, 13, 809-830.
19. Gundersen, O. E., & Kjensmo, S. (2018). State of the art: Reproducibility in artificial intelligence. In *Proceedings of the AAAI conference on artificial intelligence* (Vol. 32, No. 1).
20. Guo, C., Pleiss, G., Sun, Y., & Weinberger, K. Q. (2017). On calibration of modern neural networks. In *International conference on machine learning* (pp. 1321-1330). PMLR.
21. Harkar, S. (2026). AI agent evaluation: From prompts to metrics. IBM.
22. Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., & Steinhardt, J. (2021). Measuring massive multitask language understanding. *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2009.03300>
23. Ito, T., Campbell, M., Horesh, L., Klinger, T., & Ram, P. (2025). Quantifying artificial intelligence through algorithmic generalization. *Nature Machine Intelligence*, 7(8), 1195-1205.
24. Jiang, H., Zhang, S., Zhu, D., Bai, Y., Truong, S. T., Yi, X., ... & Xiao, Z. (2026). AI Evaluation Should Require Standardized Item-Level Data Releases. arXiv preprint arXiv:2604.03244.
25. Jimenez, C. E., Yang, J., Wettig, A., Yao, S., Pei, K., Press, O., & Narasimhan, K. (2024). SWE-Bench: Can language models resolve real-world GitHub issues? *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/abs/2310.06770>
26. Júnior, S. H. D. L., & Sant’Anna, Y. F. D. D. (2024). SuperGLUE: The AI race.
27. Kane, M. T. (2013). Validating the interpretations and uses of test scores. *Journal of educational measurement*, 50(1), 1-73.

28. Kehkashan, T., Abdullah, M., Al-Shamayleh, A. S., Ivković, N., Ismail, N. A., Ahmad, S. S. S., ... & Akhunzada, A. (2026). From benchmarks to deployment: a comprehensive review of agentic AI evaluation. *Artificial Intelligence Review*.
29. Keller, D., Kwegyir-Aggrey, K., Steed, R., Rao, A. K., Sharp, J. L., & Bergman, A. S. (2026). Expanding the AI Evaluation Toolbox with Statistical Models. In *Forty-third International Conference on Machine Learning*.
30. Khalafyan, E. A. (2025). Better Thinking or a Bigger Model? Thinking–Answering Shuffles with Qwen3 on GPQA. *Mathematical Problems of Computer Science*, 64, 17-28.
31. Kiela, D., Bartolo, M., Nie, Y., Kaushik, D., Geiger, A., Wu, Z., ... & Williams, A. (2021). Dynabench: Rethinking benchmarking in NLP. In *Proceedings of the 2021 conference of the North American chapter of the Association for Computational Linguistics: human language technologies* (pp. 4110-4124).
32. Kim, S., Joo, S., Kim, D., Jang, J., Ye, S., Shin, J., & Seo, M. (2023). The cot collection: Improving zero-shot and few-shot learning of language models via chain-of-thought fine-tuning. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing* (pp. 12685-12708).
33. Koh, P. W., Sagawa, S., Marklund, H., Xie, S. M., Zhang, M., Balsubramani, A., ... & Liang, P. (2021). Wilds: A benchmark of in-the-wild distribution shifts. In *International conference on machine learning* (pp. 5637-5664). PMLR.
34. Lambert, N. (2025). Reinforcement learning from human feedback. *arXiv preprint arXiv:2504.12501*.
35. LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *nature*, 521(7553), 436-444.
36. Li, J., Lu, W., Fei, H., Luo, M., Dai, M., Xia, M., ... & Wang, C. (2024b). A survey on benchmarks of multimodal large language models. *arXiv preprint arXiv:2408.08632*.
37. Li, L., Chen, G., Shi, H., Xiao, J., & Chen, L. (2024a). A survey on multimodal benchmarks: In the era of large ai models. *arXiv preprint arXiv:2409.18142*.
38. Liang, P., Bommasani, R., Lee, T., Tsipras, D., Soylu, D., Yasunaga, M., ... & Koreeda, Y. (2022). Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*.
39. Lin, S., Hilton, J., & Evans, O. (2022). Truthfulqa: Measuring how models mimic human falsehoods. In *Proceedings of the 60th annual meeting of the association for computational linguistics (volume 1: long papers)* (pp. 3214-3252).
40. Ling, Z., Liu, K., Yan, K., Yang, Y., Lin, W., Fan, T. H., ... & Chen, J. (2025). Longreason: A synthetic long-context reasoning benchmark via context expansion. *arXiv preprint arXiv:2501.15089*.

41. Liu, J., Xu, S., Liu, S., Li, Y., Liu, W., Liu, M., ... & Wang, H. (2025). JoyAgent-JDGenie: Technical Report on the GAIA. arXiv preprint arXiv:2510.00510.
42. Liu, T., Wang, A. X., Panescu, A., Chen, L. X., Long, W., Wei, X., ... & Zhao, H. (2026). Benchmarking AI Agents for Addressing Scientific Challenges Across Scales. arXiv preprint arXiv:2606.12736.
43. Lu, P., Bansal, H., Xia, T., Liu, J., Li, C., Hajishirzi, H., ... & Gao, J. (2024). Mathvista: Evaluating mathematical reasoning of foundation models in visual contexts. In International Conference on Learning Representations (Vol. 2024, pp. 23439-23554).
44. Lu, P., Mishra, S., Xia, T., Qiu, L., Chang, K. W., Zhu, S. C., ... & Kalyan, A. (2022). Learn to explain: Multimodal reasoning via thought chains for science question answering. *Advances in neural information processing systems*, 35, 2507-2521.
45. Lu, P., Zhang, S., Hou, Y., Ye, L., Huang, C., Chen, Z., ... & Yang, M. H. (2026). ProjDevBench: Benchmarking AI Coding Agents on End-to-End Project Development. arXiv preprint arXiv:2602.01655.
46. Masry, A., Tan, J. Q., Joty, S., & Hoque, E. (2022). Chartqa: A benchmark for question answering about charts with visual and logical reasoning. In *Findings of the association for computational linguistics: ACL 2022* (pp. 2263-2279).
47. McGregor, S., Tashev, V., Foundjem, A., Ramasethu, A., Kazemi Zarkouei, S. A., Knotz, C., ... & Frase, H. (2026). Risk management for mitigating benchmark failure modes: BenchRisk. *Advances in Neural Information Processing Systems*, 38.
48. McIntosh, T. R., Susnjak, T., Arachchilage, N., Liu, T., Xu, D., Watters, P., & Halgamuge, M. N. (2025). Inadequacies of large language model benchmarks in the era of generative artificial intelligence. *IEEE Transactions on Artificial Intelligence*.
49. Menzies, T., Turhan, B., Bener, A., Gay, G., Cukic, B., & Jiang, Y. (2008). Implications of ceiling effects in defect predictors. In *Proceedings of the 4th international workshop on Predictor models in software engineering* (pp. 47-54).
50. Messick, S. (1989). Validity. In R. L. Linn (Ed.), *Educational measurement* (3rd ed., pp. 13-104). New York, NY: American Council on education and Macmillan.
51. Minderer, M., Djolonga, J., Romijnders, R., Hubis, F., Zhai, X., Houlsby, N., ... & Lucic, M. (2021). Revisiting the calibration of modern neural networks. *Advances in neural information processing systems*, 34, 15682-15694.
52. Mirzadeh, I., Alizadeh-Vahid, K., Shahrokhi, H., Tuzel, O., Bengio, S., & Farajtabar, M. (2025). Gsm-symbolic: Understanding the limitations of mathematical reasoning in large language models. In *International Conference on Learning Representations* (Vol. 2025, pp. 94743-94765).

53. Miyai, A., Yang, J., Zhang, J., Ming, Y., Lin, Y., Yu, Q., ... & Aizawa, K. (2024). Generalized out-of-distribution detection and beyond in vision language model era: A survey. arXiv preprint arXiv:2407.21794.
54. Mohajan, H. K. (2025). Artificial Intelligence: Prospects and challenges in future progression. *Art and Society*, 4(7), 38-50.
55. Morishige, M., & Koshihara, R. (2025). Ensuring Reproducibility in Generative AI Systems for General Use Cases: A Framework for Regression Testing and Open Datasets. arXiv preprint arXiv:2505.02854.
56. Mundlamuri, R., Gunnam, G. R., Mysari, N. K., & Pujuri, J. (2025). The Evolution of AI: From Classical Machine Learning to Modern Large Language Models. *Ieee Access*.
57. National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0) (NIST AI 100-1).
58. Nunnally, J.C. and Bernstein, I.H. (1994) The Assessment of Reliability. *Psychometric Theory*, 3, 248-292.
59. Pandit, K. (2026). Beyond the Scaling Ceiling: An Architectural Phase Transition in AI Reliability.
60. Phan, L., Gatti, A., Han, Z., Li, N., Hu, J., Zhang, H., ... & Wykowski, J. (2025). Humanity's last exam. arXiv preprint arXiv:2501.14249.
61. Pineau, J., Vincent-Lamarre, P., Sinha, K., Larivière, V., Beygelzimer, A., d'Alché-Buc, F., ... & Larochelle, H. (2021). Improving reproducibility in machine learning research (a report from the neurips 2019 reproducibility program). *Journal of machine learning research*, 22(164), 1-20.
62. Prokopchina, S. V. (2024). A New Direction in Artificial Intelligence: Measuring Artificial Intelligence. In *2024 XXVII International Conference on Soft Computing and Measurements (SCM)* (pp. 3-6). IEEE.
63. Raji, I. D., Smart, A., White, R. N., Mitchell, M., Gebru, T., Hutchinson, B., ... & Barnes, P. (2020). Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In *Proceedings of the 2020 conference on fairness, accountability, and transparency* (pp. 33-44).
64. Rein, D., Hou, B. L., Stickland, A. C., Petty, J., Pang, R. Y., Dirani, J., ... & Bowman, S. R. (2023). Gpqa: A graduate-level google-proof q&a benchmark. arXiv preprint arXiv:2311.12022.
65. Reuel, A., Hardy, A., Smith, C., Lamparth, M., Hardy, M., & Kochenderfer, M. J. (2024). Betterbench: Assessing ai benchmarks, uncovering issues, and establishing best practices. *Advances in Neural Information Processing Systems*, 37, 21763-21813.

66. Reuel, A., Hardy, A., Smith, C., Lamparth, M., Hardy, M., & Kochenderfer, M. J. (2024b). What Makes a Good AI Benchmark?. Policy Brief, Stanford University Human-Centered Artificial Intelligence (HAI), December.
67. Russell, S., & Norvig, P. (2020). Artificial intelligence: a modern approach, 4/E.
68. Salaudeen, O., Reuel, A., Ahmed, A., Bedi, S., Robertson, Z., Sundar, S., ... & Koyejo, S. (2025). Measurement to meaning: A validity-centered framework for ai evaluation. arXiv preprint arXiv:2505.10573.
69. Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., ... & Dennison, D. (2015). Hidden technical debt in machine learning systems. *Advances in neural information processing systems*, 28.
70. Shen, Y., Song, K., Tan, X., Zhang, W., Ren, K., Yuan, S., ... & Zhuang, Y. (2024). Taskbench: Benchmarking large language models for task automation. *Advances in Neural Information Processing Systems*, 37, 4540-4574.
71. Shukla, M. (2025). Evaluation and Benchmarking of Generative and Agentic AI Systems: A Comprehensive Survey. Available at SSRN 5927324.
72. Sierra-Herrera, O. H., Niño, M. E. G., Correa, E. F. C., Leon-Medina, J. X., & Pozo, F. (2026). A Reproducible benchmarking methodology for machine learning hardware: Performance–energy trade-offs from GPUs to Apple Silicon. *Algorithms*, 19(5), 363.
73. Snyder, H. (2019). Literature review as a research methodology: An overview and guidelines. *Journal of business research*, 104, 333-339.
74. Solanke, A. A. (2026). AISafetyBenchExplorer: A Metric-Aware Catalogue of AI Safety Benchmarks Reveals Fragmented Measurement and Weak Benchmark Governance. arXiv preprint arXiv:2604.12875.
75. Srivastava, A., Rastogi, A., Rao, A., Shoeb, A. A. M., Abid, A., Fisch, A., ... & Wang, G. X. (2023). Beyond the imitation game: Quantifying and extrapolating the capabilities of language models. *Transactions on machine learning research*.
76. Starace, G., Jaffe, O., Sherburn, D., Aung, J., Chan, J. S., Maksin, L., ... & Patwardhan, T. (2025). PaperBench: Evaluating AI's Ability to Replicate AI Research. arXiv preprint arXiv:2504.01848.
77. Stolovitzky, G., Saez-Rodriguez, J., Bletz, J., Albrecht, J., Andreoletti, G., Costello, J. C., & Boutros, P. (2023). AI Competitions and Benchmarks: The life cycle of challenges and benchmarks. arXiv preprint arXiv:2312.05296.
78. Stoyanovich, J., & Howe, B. (2019). Nutritional labels for data and models. *A Quarterly bulletin of the Computer Society of the IEEE Technical Committee on Data Engineering*, 42(3).

79. Suzgun, M., Scales, N., Schärli, N., Gehrmann, S., Tay, Y., Chung, H. W., ... & Wei, J. (2023). Challenging big-bench tasks and whether chain-of-thought can solve them. In Findings of the Association for Computational Linguistics: ACL 2023 (pp. 13003-13051).
80. Thomas, R., & Uminsky, D. (2020). The problem with metrics is a fundamental problem for AI. arXiv preprint arXiv:2002.08512.
81. Wang, A., Pruksachatkun, Y., Nangia, N., Singh, A., Michael, J., Hill, F., ... & Bowman, S. (2019). Superglue: A stickier benchmark for general-purpose language understanding systems. *Advances in neural information processing systems*, 32.
82. Wang, Y., Ma, X., Zhang, G., Ni, Y., Chandra, A., Guo, S., ... & Chen, W. (2024). Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *Advances in Neural Information Processing Systems*, 37, 95266-95290.
83. Webster, J., & Watson, R. T. (2002). Analyzing the past to prepare for the future: Writing a literature review. *MIS quarterly*, 26(2), xiii-xxiii.
84. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., ... & Zhou, D. (2022). Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems*, 35, 24824-24837.
85. Wei, T., Li, T. W., Liu, Z., Ning, X., Yang, Z., Zou, J., ... & He, J. (2026). Agentic reasoning for large language models. arXiv preprint arXiv:2601.12538.
86. Winfield, A. F., & Studley, M. (2024). On the relationship between Benchmarking, Standards and Certification in Robotics and AI. In *Producing Artificial Intelligent Systems: The Roles of Benchmarking, Standardisation and Certification* (pp. 1-9). Cham: Springer Nature Switzerland.
87. Yeung, K. (2020). Recommendation of the council on artificial intelligence (OECD). *International legal materials*, 59(1), 27-34.
88. Yu, Y., Zhang, Q. W., Qiao, L., Yin, D., Li, F., Wang, J., ... & Sun, X. (2025). Sequential-niah: A needle-in-a-haystack benchmark for extracting sequential needles from long contexts. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (pp. 29438-29456).
89. Yu, Z., Zhao, Y., Cohan, A., & Zhang, X. P. (2024). Humaneval pro and mbpp pro: Evaluating large language models on self-invoking code generation. arXiv preprint arXiv:2412.21199.
90. Zhai, W., Wang, Z., Wang, J., Yang, B., Li, X., Xu, X., ... & Zhao, B. (2026). HLE-Verified: A Systematic Verification and Structured Revision of Humanity's Last Exam. arXiv preprint arXiv:2602.13964.
91. Zhang, Z., Zhang, A., Li, M., & Smola, A. (2022). Automatic chain of thought prompting in large language models. arXiv preprint arXiv:2210.03493.

92. Zhao, Y., Wei, T., Zou, J., Ning, X., Bei, Y., Chen, L., ... & He, J. (2026). PAPERMIND: Benchmarking Agentic Reasoning and Critique over Scientific Papers in Multimodal LLMs. In Findings of the Association for Computational Linguistics: ACL 2026 (pp. 10457-10474).